

ARCHITECTURE STUDY | XR-ACCEL-01 | REFERENCE: VECTOR CPU EXTENSION

CLAIM UNDER REVIEW The shared-memory SoC block improves XR efficiency over a vector CPU extension and a loosely coupled accelerator within 3 W, preserving QoS and deployability.

Intent

Improve mobile XR efficiency without violating power, QoS, memory, or deployment constraints.

Task

Compare vector CPU extension (reference), loosely coupled accelerator, and shared-memory SoC block.

Design space

Legal accelerator, memory, and vector choices; software redesign and full SoC changes are out.

Representation

XRbench traces, architecture parameters, compiler assumptions, power model, and uncertainty record.

Environment

Simulator or cost model plus workload harness; the contract defines actions and observations.

Method role

Generate candidates, predict cost, optimize, challenge assumptions, and explain mechanisms.

Feedback budget

Many cheap proxies, fewer simulations, and two high-fidelity checks before claim review.

Evidence

Links to runs, three-way comparisons, sensitivities, provenance, and limits in the evidence record.

Failed runs / rejected alternatives

Links to power violations, QoS misses, bandwidth overruns, proxy mismatches, and discarded options.

Rejection checks and authority

Power, QoS, proxy, and cross-tool checks may block or weaken the claim; the architect decides what follows.

What the evidence supports

The evidence supports higher-fidelity evaluation only; no RTL, integration, or product decision.

Accountable decision

The named architect owns scope changes, waivers, escalation, and the decision about what happens next.

The card indexes the study, claim, artifacts, evidence, and decision. It does not replace the records.